Khushboo K. Rao (she/her)

Postdoctoral Researcher, Institute of Astronomy, National Central University, Taiwan

khushboo@astro.ncu.edu.tw | ADS library | ORCID 0000-0001-7470-9192 | LinkedIn

RESEARCH PROFILE

I work on how stars in open clusters depart from canonical evolutionary pathways, both individually and within stellar clusters. My research focuses on uncovering the physical mechanisms behind unusual stellar populations, such as blue stragglers, blue lurkers, and fast and slow rotating intermediate-mass stars. I combine ultraviolet and optical photometry, spectroscopy, high cadence time series and machine learning methods.

Keywords: open clusters; blue straggler and blue lurker stars; extended main sequence turnoffs; stellar rotation; Herbig Ae Be stars; X-rays; eclipsing binaries and light curve modelling; time domain surveys; machine learning

PUBLICATION RECORD

15 refereed journal articles, 4 conference proceedings and 2 preprints under review. 7 as first author and 7 as second author. h-index 10 (ADS, August 2026). Full list maintained in the ADS library linked above.

APPOINTMENTS

Nov 2023 - **Postdoctoral Researcher**, Institute of Astronomy, National Central University, Taiwan

Apr - May 2025 **Visiting Researcher**, Tata Institute of Fundamental Research, Mumbai, India

EDUCATION

2018 - 2023 **Ph.D., Observational Astrophysics**

Birla Institute of Technology and Science, Pilani, Rajasthan, India

Thesis: A study of blue straggler stars in open clusters. Supervisor: Dr Kaushar Vaidya

2015 - 2017 **M.Sc., Physics**

University College of Science, Mohanlal Sukhadia University, Udaipur, India

2012 - 2015 **B.Sc., Physics, Mathematics and Chemistry**

Meera Girls College, Mohanlal Sukhadia University, Udaipur, India

AWARDED TELESCOPE TIME

2026, P117 **VLT / GIRAFFE, 20 hours** Co-I

Constraining the mass-luminosity relation and evolution of blue stragglers with eclipsing systems in Collinder 261

Role: Target selection, proposal writing, data reduction once observed

2026A **Gemini North / MAROON-X, 36.3 hours, Bands 3 and 4** Science-PI

Deciphering the origins of yellow straggler stars in open clusters

Role: Target selection, proposal writing, data reduction once observed

2026A **Gemini / GHOST, 10.5 hours, Band 3** Science-PI

Constraining the mass-luminosity relation and evolution of blue stragglers with eclipsing systems

Role: Target selection, proposal writing, data reduction once observed

2023A **DOT / ADFOSC, 48 hours** PI

Radial velocity monitoring of eclipsing blue straggler stars in open clusters

Role: Target selection, proposal writing, data reduction in progress

2022B **HCT / HFOSC, 1 night** PI

Spectroscopic study of blue straggler stars of Berkeley 17

Role: Target selection, proposal writing, data reduction in progress

2022B **AstroSat / UVIT, 12.5 hours** Co-I

UVIT study of blue straggler stars in metal-poor open clusters

Role: Proposal writing

REFEREED PUBLICATIONS

1. **Rao, K. K.**; Chen, W. (2026). *Investigating Extended Main-Sequence Turnoffs in Galactic Open Clusters.* ApJ 1002, 103 (2026) **[first author]**
2. Jadhav, V. V.; **Rao, K. K.**; Reggiani, E. (2026). *A rotation-based census of blue lurker candidates in open clusters.* A&A 707, A275 (2026)
3. Dage, K. C.; Hunt, E. L.; Anderson-Baldwin, J.; ... ; **Rao, K. K.**; ... [14 authors, position 5] (2026). *The Secret Lives of Open Clusters: a Multiwavelength Examination of Three Open Clusters.* PASA 43, e013 (2026)
4. Chand, K.; Rao, K.; Vaidya, K.; Panthi, A. (2024). *Characterization of blue and yellow straggler stars of Berkeley 39 using Swift/UVOT.* AJ 168, 278 (2024)
5. Balan, S.; **Rao, K. K.**; Vaidya, K.; Agarwal, M.; et al. [5 authors] (2024). *Dynamical Evolution of Four Old Galactic Open Clusters traced by their constituent stars with \textit{Gaia} DR3.* AJ 168, 204 (2024)
6. **Rao, K. K.**; Vaidya, K.; Agarwal, M.; Balan, S.; et al. [5 authors] (2023). *Determination of dynamical ages of open clusters through the A+ parameter (Paper II).* MNRAS 526, 1057-1074 (2023) **[first author]**
7. **Rao, K. K.**; Bhattacharya, S.; Vaidya, K.; Agrawal, M. (2022). *Discovery of double BSS sequences in the old Galactic open cluster Berkeley 17.* MNRAS: Letters 518, L7-L12 (2022) **[first author]**
8. Bhattacharya, S.; **Rao, K. K.**; Agarwal, M.; Balan, S.; et al. [5 authors] (2022). *A Gaia EDR3 search for tidal tails in disintegrating open clusters.* MNRAS 517, 3525-3549 (2022)
9. Panthi, A.; Vaidya, K.; Jadhav, V.; **Rao, K. K.**; et al. [7 authors] (2022). *UOCS VIII. Ultraviolet study of the open cluster NGC 2506 using ASTROSAT.* MNRAS 516, 5318-5330 (2022)
10. **Rao, K. K.**; Vaidya, K.; Agarwal, M.; Panthi, A.; et al. [6 authors] (2022). *Characterization of hot populations of Melotte 66 open cluster using Swift/UVOT.* MNRAS 516, 2444-2454 (2022) **[first author]**
11. Vaidya, K.; Panthi, A.; Agarwal, M.; ... ; **Rao, K. K.**; ... [7 authors, position 5] (2022). *UOCS VII. Blue Straggler Populations of Open Cluster NGC 7789 with UVIT/AstroSat.* MNRAS 511, 2274-2284 (2022)
12. Bhattacharya, S.; Agarwal, M.; **Rao, K. K.**; Vaidya, K. (2021). *Tidal tails in the disintegrating open cluster NGC 752.* MNRAS 505, 1607-1613 (2021)
13. **Rao, K. K.**; Vaidya, K.; Agarwal, M.; Bhattacharya, S. (2021). *Determination of dynamical ages of open clusters through the A+ parameter (Paper I).* MNRAS 508, 4919-4937 (2021) **[first author]**
14. Agarwal, M.; **Rao, K. K.**; Vaidya, K.; Bhattacharya, S. (2020). *ML-MOC: Machine Learning (kNN and GMM) based Membership Determination for Open Clusters.* MNRAS 502, 2582-2599 (2021)
15. Vaidya, K.; **Rao, K. K.**; Agarwal, M.; Bhattacharya, S. (2020). *Blue Straggler Populations of Seven Open Clusters with Gaia DR2.* MNRAS 496, 2402-2421 (2020)

PREPRINTS UNDER REVIEW

1. **Rao, K. K.**; Banerjee, S. (2026). *Examining the stellar-merger origin of the blue main sequence in the open cluster NGC 3532 with N-body simulations.* submitted, arXiv:2607.18681 **[first author]**
2. **Rao, K. K.**; et al. Uncovering angular momentum loss in intermediate-mass stars in the star-forming cluster Trumpler 14 **[first author]**

CONFERENCE PROCEEDINGS

1. **Rao, K. K.**; Chen, W. (2025). *Investigating Unusually Extended Main Sequence Turnoff of the Galactic Open Cluster NGC 3532. IAU MODEST conference,* submitted, arXiv:2512.05458
2. **Rao, K. K.**; et al. (2024). *Blue straggler stars of NGC 7789 and NGC 2506 using AstroSat/UVIT.* Bulletin de la Societe Royale des Sciences de Liege
3. **Rao, K. K.**; et al. (2024). *Blue straggler stars: setting up a dynamical clock for open clusters.* Bulletin de la Societe Royale des Sciences de Liege
4. Chand, K.; **Rao, K. K.**; Vaidya, K.; Panthi, A. (2023). *Hot Stellar Populations of Berkeley 39 using Swift/UVOT.* Bulletin de la Société Royale des Sciences de Liège, 262-272 (2024)

INVITED TALKS AND COLLOQUIA

August 2026	*Blue straggler stars in the big-data era: the role of machine-learning*. Astro-AI Seminar, Center for Astrophysics, Harvard & Smithsonian, Cambridge, USA
Oct 2025	*Beyond a single isochrone: how blue stragglers and extended main sequence turnoffs reshape our view of stellar evolution*. Colloquium, National Tsing Hua University, Hsinchu, Taiwan
May 2025	*The formation and role of blue straggler stars in open clusters*. Colloquium, Tata Institute of Fundamental Research, Mumbai, India
Jan 2025	*The formation and role of blue straggler stars in open clusters*. LSST Stars, Milky Way and Local Volume collaboration meeting, online
Dec 2023	*Investigating dynamical ages of open clusters using blue straggler stars*. Colloquium, National Central University, Zhongli, Taiwan
Nov 2024	*Star clusters*. Invited talk for school students, Kinmen Senior High School, Kinmen, Taiwan

CONTRIBUTED TALKS

May 2026	*Unravelling formation mechanisms of stragglers in open clusters*. 44th Meeting of the Astronomical Society of India
May 2026	*Early spin down pathways of intermediate-mass stars in the star-forming cluster Trumpler 14*. Asia-Pacific Regional IAU Meeting, Hong Kong
Feb 2026	*Role of high cadence TAOS II time series photometric data in the variability study of open clusters and their exotic populations*. TAOS II Workshop, National Central University, Taiwan
Jan 2026	*When fast turns slow: unravelling the rotation puzzle in the young open cluster Trumpler 14*. Annual Meeting of the Physical Society of Taiwan, National Chung Cheng University
Jan 2025	*Investigating unusually extended main sequence turnoffs in Galactic open clusters*. Annual Meeting of the Physical Society of Taiwan, National Sun Yat-sen University
May 2024	*Detection of hot white dwarf companions of blue stragglers and blue lurkers in Berkeley 31 and Melotte 66*. Scientific Assembly of the Astronomical Society of the Republic of China

POSTERS

2025	IAU MODEST-25, Seoul National University, South Korea
2024	MODEST-24, Nicolaus Copernicus Astronomical Center, Warsaw, Poland
2024	3rd BINA Workshop, Graphic Era Hill University, Bhimtal, India
2022	Gaia Symposium: DR3 and Beyond, Indian Institute of Astrophysics, Bengaluru
2019-2022	37th to 40th Annual Meetings of the Astronomical Society of India

TEACHING AND MENTORING

2025	**Lecturer**, International School of Young Astronomers

2024 - 2026	**Lecturer**, Summer Student Camp, Institute of Astronomy, National Central University. Introductory astronomy and Python programming, three cohorts
2018 - 2023	**Research mentor**, BITS Pilani. Supervised three undergraduate and two master's students on open cluster projects
2018 - 2023	**Laboratory instructor**, BITS Pilani. First year B.Tech physics laboratory

SERVICE AND PROFESSIONAL MEMBERSHIP

2026	**Journal Club Coordinator**, Institute of Astronomy, National Central University. Convene weekly sessions for graduate students and researchers
Current	**Time Allocation Committee member**, Lulin Observatory, National Central University
Current	**Member**, LSST Stars, Milky Way and Local Volume Science Collaboration
Current	**Reviewer**, Astronomy & Astrophysics (A&A) and Journal of Astronomy and Astrophysics

TECHNICAL SKILLS

Observing	Spectroscopic and photometric observations with 1 m and 4 m class telescopes. Proposal writing as PI and Co-I.
Analysis	Photometry, spectroscopy, light curve modelling, spectral energy distribution fitting, time series and variability analysis, machine learning applied to cluster membership (kNN and Gaussian mixture models).
Data	Gaia, AstroSat UVIT, Swift UVOT, TESS, Photometry, VLT and Gemini spectroscopy.
Computing	Python, R, ADQL, HTML and CSS.

AWARDS AND QUALIFICATIONS

2018	**UGC National Eligibility Test** for Assistant Professorship, All India Rank 146
2018	**Graduate Aptitude Test in Engineering (GATE)**, qualified
2012 - 2017	**Ministry of Human Resource Development scholarship**, India, undergraduate and master's study

REFERENCES

Prof. Wen-Ping Chen. Professor, Institute of Astronomy, National Central University, Taiwan. wchen@astro.ncu.edu.tw

Dr Kaushar Vaidya. Associate Professor, doctoral supervisor, Department of Physics, BITS Pilani, India. kaushar@pilani.bits-pilani.ac.in

Dr Sambaran Banerjee. Independent researcher and principal investigator, HISKP and AIfA, University of Bonn, Germany. sambaran.banerjee@gmail.com